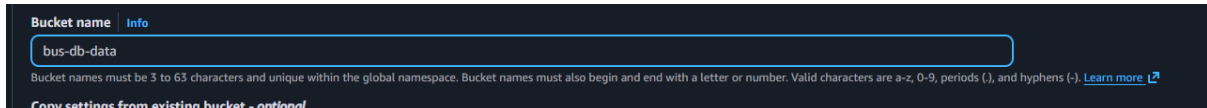


Building the app on AWS:

S3 bucket:

A bucket was created to store the schedules in json format.

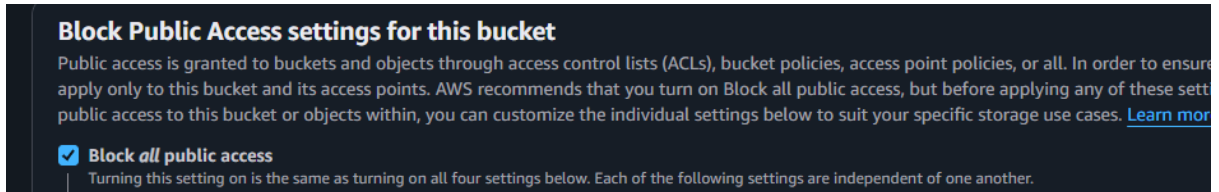


Bucket name [Info](#)

bus-db-data

Bucket names must be 3 to 63 characters and unique within the global namespace. Bucket names must also begin and end with a letter or number. Valid characters are a-z, 0-9, periods (.), and hyphens (-). [Learn more](#) [?](#)

Copy settings from existing bucket - optional



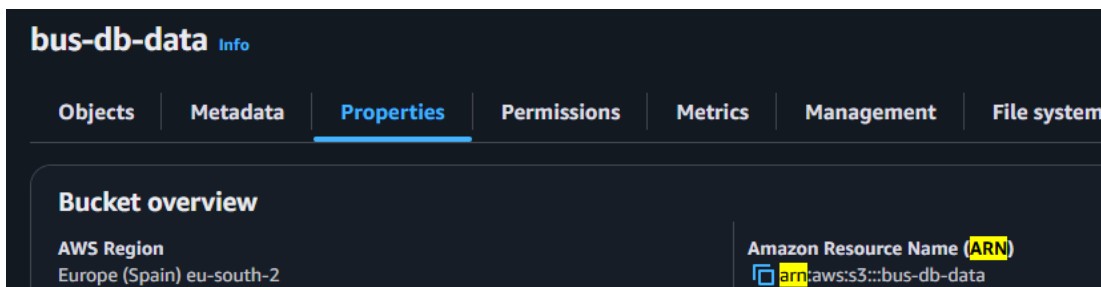
Block Public Access settings for this bucket

Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access is applied only to this bucket and its access points, AWS recommends that you turn on Block all public access, but before applying any of these settings to this bucket or objects within, you can customize the individual settings below to suit your specific storage use cases. [Learn more](#) [?](#)

☒ **Block all public access**

Turning this setting on is the same as turning on all four settings below. Each of the following settings are independent of one another.

We'll also need its ARN to manage security and permissions (as well as debug some lambda issues)



bus-db-data [Info](#)

[Objects](#) [Metadata](#) [Properties](#) [Permissions](#) [Metrics](#) [Management](#) [File systems](#)

Bucket overview

AWS Region Europe (Spain) eu-south-2	Amazon Resource Name (ARN)  arn:aws:s3:::bus-db-data
--	---

DynamoDB:

A DynamoDB database was created with PK and SK field names.

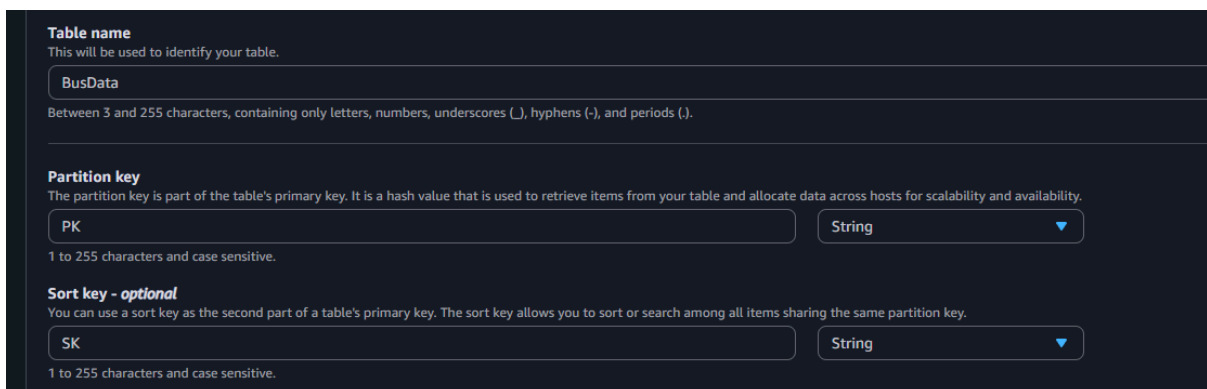


Table name

This will be used to identify your table.

BusData

Between 3 and 255 characters, containing only letters, numbers, underscores (_), hyphens (-), and periods (.).

Partition key

The partition key is part of the table's primary key. It is a hash value that is used to retrieve items from your table and allocate data across hosts for scalability and availability.

PK String

1 to 255 characters and case sensitive.

Sort key - optional

You can use a sort key as the second part of a table's primary key. The sort key allows you to sort or search among all items sharing the same partition key.

SK String

1 to 255 characters and case sensitive.

Table settings

☐ Default settings
The fastest way to create your table. You can modify most of these settings after your table has been created. To modify these settings now, choose 'Customize settings'.

☒ Customize settings
Use these advanced features to make DynamoDB work better for your needs.

Table class [info](#)
Select table class to optimize your table's cost based on your workload requirements and data access patterns.

Choose table class

☒ **DynamoDB Standard**
The default general-purpose table class. Recommended for the vast majority of tables that store frequently accessed data, with throughput (reads and writes) as the dominant table cost.

☐ **DynamoDB Standard-IA**
Recommended for tables that store data that is infrequently accessed, with storage as the dominant table cost.

► Capacity calculator [info](#)

Read/write capacity settings [info](#)

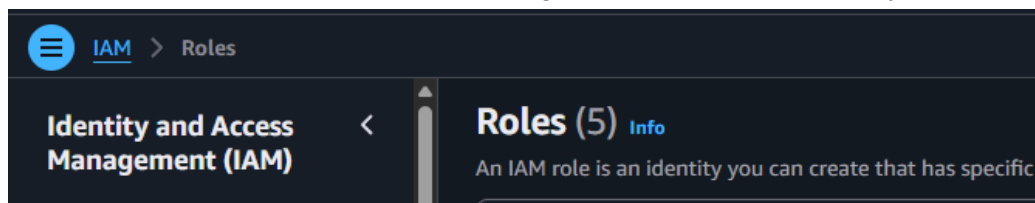
Capacity mode

☒ **On-demand**
Simplify billing by paying for the actual reads and writes your application performs.

☐ **Provisioned**
Manage and optimize your costs by allocating read/write capacity in advance.

IAM:

IAM Roles were needed in order to manage permissions and security.



Two different roles for Lambda:

- A role for writing to the database and updating it
- A role for querying the database that's read-only.

In order to know exactly which permissions were needed, lambda methods had to be coded first. And there was also trial and error due to dependencies.

In order to create those roles:

- We select specific the service

Select trusted entity Info

Trusted entity type

☒ **AWS service**
 Allow AWS services like EC2, Lambda, or others to perform actions in this account.

☐ **AWS account**
 Allow entities in other AWS accounts belonging to you or a 3rd party to perform actions in this account.

☐ **SAML 2.0 federation**
 Allow users federated with SAML 2.0 from a corporate directory to perform actions in this account.

☐ **Custom trust policy**
 Create a custom trust policy to enable others to perform actions in this account.

Use case

Allow an AWS service like EC2, Lambda, or others to perform actions in this account.

Service or use case

Lambda

Choose a use case for the specified service.

Use case

☒ **Lambda**
 Allows Lambda functions to call AWS services on your behalf.

- We attach relevant policies

Add permissions Info

Permissions policies (1/1149) Info

Choose one or more policies to attach to your new role.

×

Filter by Type
 All types

☐	Policy name ↗	Type
☐	AWSLambdaBasicDurableExecutionRolePolicy	AWS managed
☒	AWSLambdaBasicExecutionRole	AWS managed

- We select a name and save the role

Role name

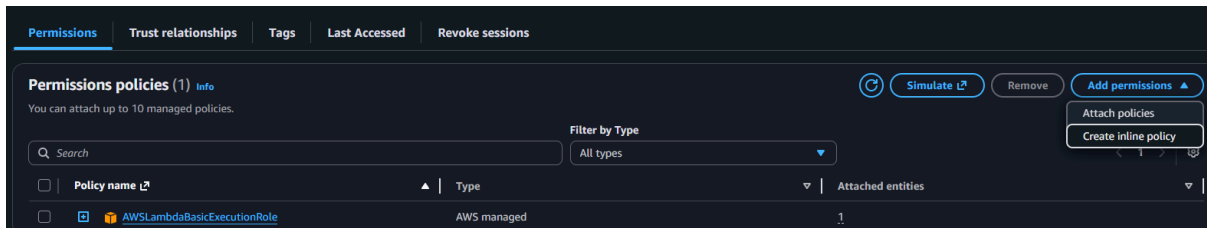
Enter a meaningful name to identify this role.

import-from-lambda-role

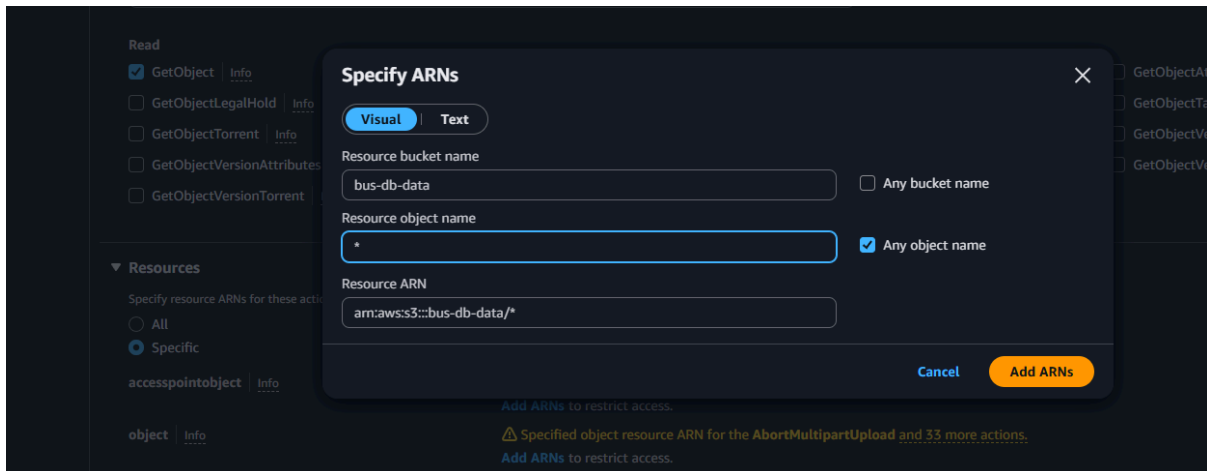
Maximum 64 characters. Use alphanumeric and '+ = , @ - _ ' characters.

Description

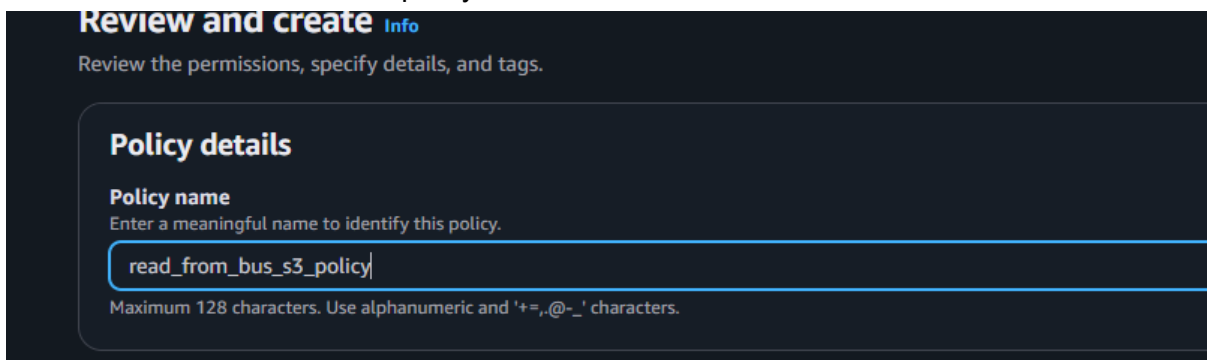
- We enter the role section and add more specific policies.



- with the S3 ARN code, we specify related resources the policy allows to access



- we name and save the policy



We add another policy that allows writing on DynamoDB, selecting methods used on the Lambda import method at first. Later, when Lambda is actually tested we'll add any missing permissions. In this case "DescribeTable"

Manual actions | [Add actions](#)

☐ All DynamoDB actions (dynamodb:*)

Access level

► **List (6)**

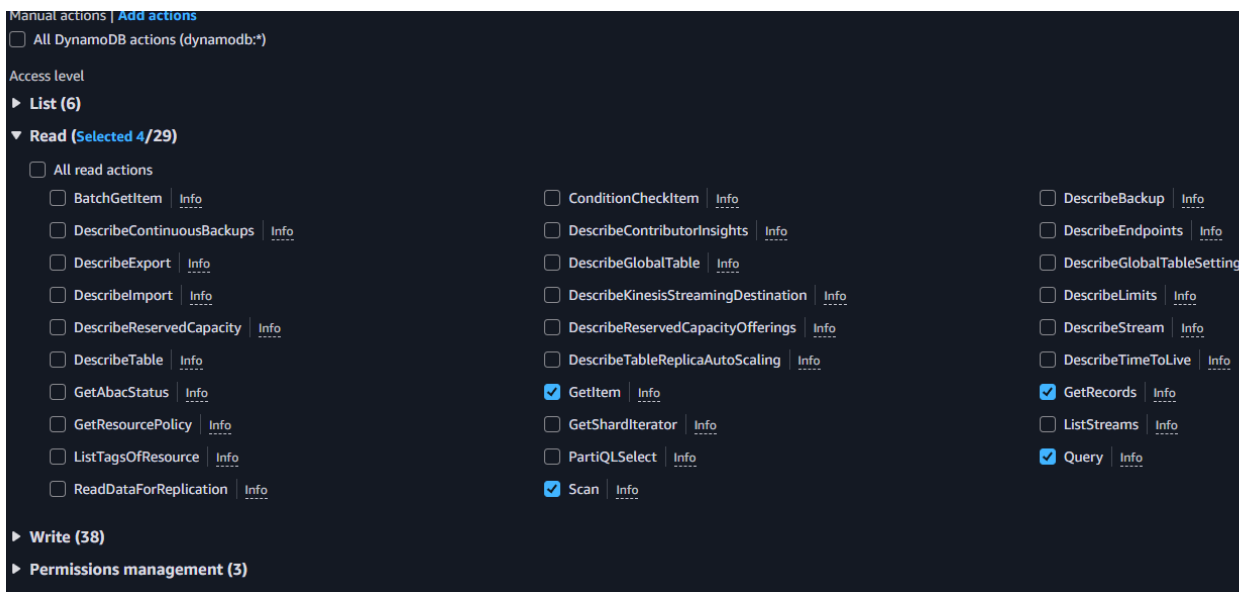
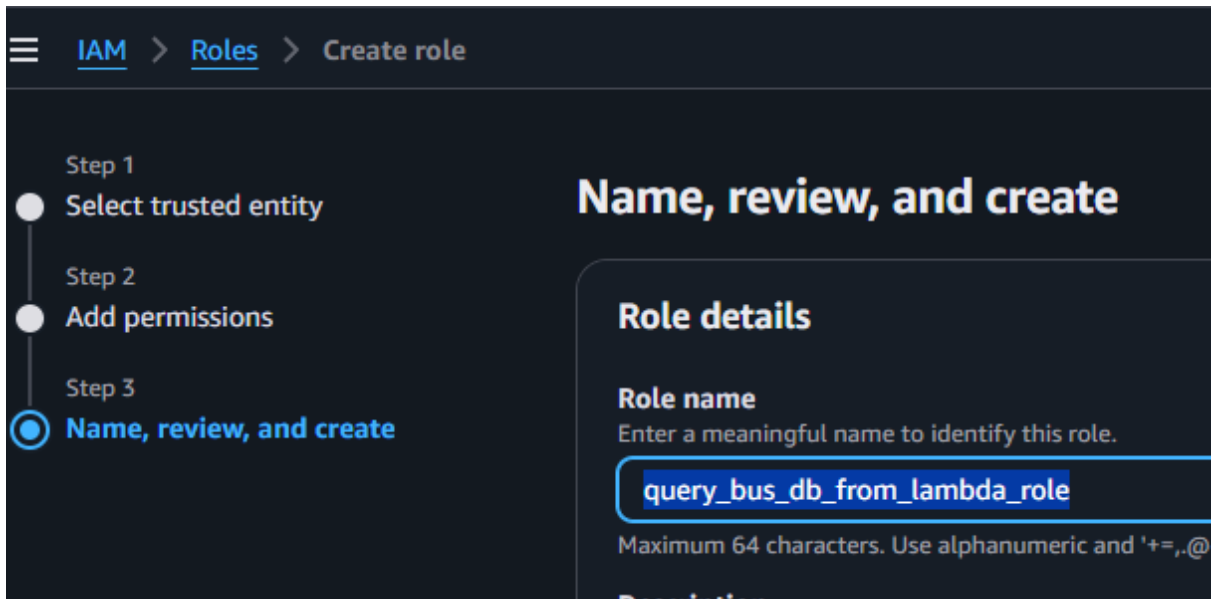
► **Read (29)**

▼ **Write (Selected 2/38)**

☐ All write actions

☐ AssociateTableReplica Info	☒ BatchWriteItem Info
☐ CreateGlobalTable Info	☐ CreateGlobalTableWitness Info
☐ CreateTableReplica Info	☐ DeleteBackup Info
☐ DeleteItem Info	☐ DeleteTable Info
☐ DisableKinesisStreamingDestination Info	☐ EnableKinesisStreamingDestination Info
☐ ImportTable Info	☐ InjectError Info
☐ PartiQLInsert Info	☐ PartiQLUpdate Info
☒ PutItem Info	☐ ReplicateSettings Info

- Similarly, a role for reading the database is created



- Later, manual edition of policies was also used

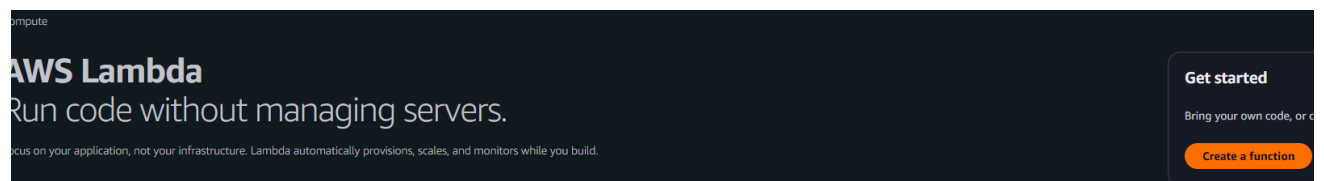
```
Policy editor
1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Sid": "VisualEditor0",
6       "Effect": "Allow",
7       "Action": [
8         "s3:GetObject",
9         "s3:ListBucket"
10      ],
11      "Resource": [
12        "arn:aws:s3:::bus-db-data",
13        "arn:aws:s3:::bus-db-data/*"
14      ]
15    }
16  ]
17 }
```

Lambda methods:

For the number of requests expected, and the lightweight processes involved, lambda was considered a good option, simple and cost effective.

We use 3 methods, all based on Python. For each one:

- We choose "create a function"

The image shows the top section of the AWS Lambda console landing page. It has a dark blue background. On the left, the text "compute" is partially visible. The main heading is "AWS Lambda" in white, followed by the tagline "Run code without managing servers." in a smaller font. Below this, a line of text reads "Focus on your application, not your infrastructure. Lambda automatically provisions, scales, and monitors while you build." On the right side, there is a "Get started" button with the text "Bring your own code, or choose from a selection of templates." and a prominent orange "Create a function" button.

- We select Language and IAM role for the Lambda function

Basic information

Function name
Enter a name that describes the purpose of your function.

import_bus_db

Function name must be 1 to 64 characters, must be unique to the Region, and can't include spaces. Valid characters are a-z, A-Z, 0-9, hyphens (-), and underscores (_).

Runtime [Info](#)
Choose the language to use to write your function. Note that the console code editor supports only Node.js, Python, and Ruby.

Python 3.14

Permissions
By default, Lambda will create an execution role with permissions to upload logs to Amazon CloudWatch Logs. To use a different role, choose **Custom execution role** under **More settings**.

Custom settings [Info](#)

Durable execution - [new](#) [?](#) ☐
Build resilient, stateful applications with automatic failure recovery.

EC2 capacity provider - [new](#) [?](#) ☐
Run Lambda on your preferred EC2 instance types instead of Lambda's serverless compute (default).

Additional settings [?](#)
Customize your function architecture, execution role, HTTP(S) endpoints, tenant isolation mode, VPC, code signing, KMS key, and tags.

General [Info](#)

Execution role
To access other AWS services and resources your function needs an IAM role. Choose a role with the right permissions for your function.

Choose an existing role

Q |

Lambda function service-assumable roles

- import-from-lambda-role
- query_bus_db_from_lambda_role

Additional roles

- AWSServiceRoleForResourceExplorer
- AWSServiceRoleForSupport
- AWSServiceRoleForTrustedAdvisor
- ImageGeneratorBucketUploader
- TestRoleEC2

Custom execution role ☒
Use custom execution role instead of new role with basic Lambda permissions (default).

- We update code from a zip file, which has all needed code prepared to be uploaded directly

Code source [Info](#)

Open In Visual Studio Code [?](#) [Update](#) [?](#)

Update from a .zip file [?](#)

Update from a file in Amazon S3 location [?](#)

Start

EXPLORER

- IMPORT_BUS_DB
 - lambda_function.py

lambda_function.py

1 import json

2

Amazon Q Tip 1/3: Start typing to get suggestions ([ESC] to exit)

- We select the handler in the settings, identifying one single method within the zip file system

Edit runtime settings

Runtime settings [Info](#)

Runtime
Choose the language to use to write your function. Note that the console code editor supports only Node.js, Python, and Ruby.

Python 3.14

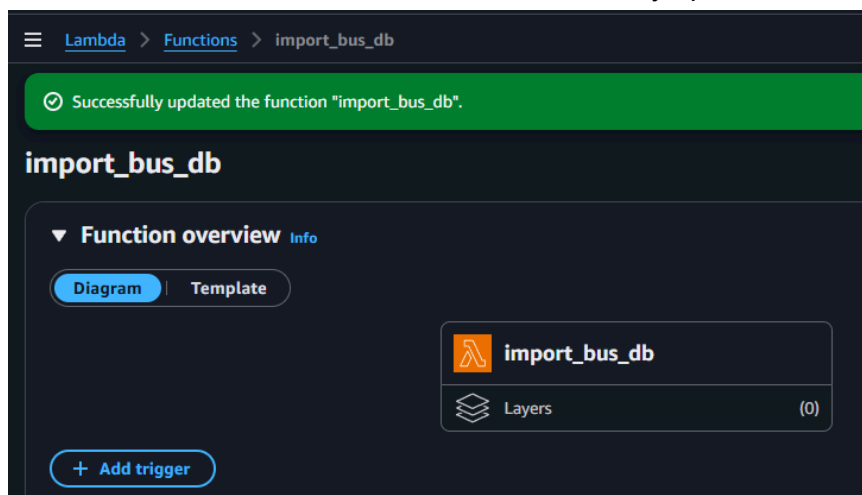
Handler [Info](#)

import_bus_db.lambda_handler

- Whenever we update the code, we also hit the “deploy” button (image below was a first draft of the code, not the final one)


```
import_bus_db.py
17 # manages retries automatically and may be faster than naive put_item in a loop into table directly
18 with table.batch_writer() as batch:
19     for route in data["routes"]:
20         for trip in route["trips"]:
21             for stop in trip["stops"]:
22                 batch.put_item(Item={
23                     "PK": "CONFIG",
24                     "SK": "ACTIVE_DATASET",
25                     "route": route,
26                     "trip": trip,
27                     "stop": stop
28                 })
29
30 ##### 4. Update active dataset "information"
31
32 table.put_item(Item={
33     "PK": "CONFIG",
34     "SK": "ACTIVE_DATASET",
35     "route": route,
36     "trip": trip,
37     "stop": stop
38 })
```

- In the import Lambda function, we have to add a trigger to S3 too, so uploading a schedule to the S3 bucket will automatically update the database



- The trigger will happen for every .json file uploaded.

Trigger configuration Info

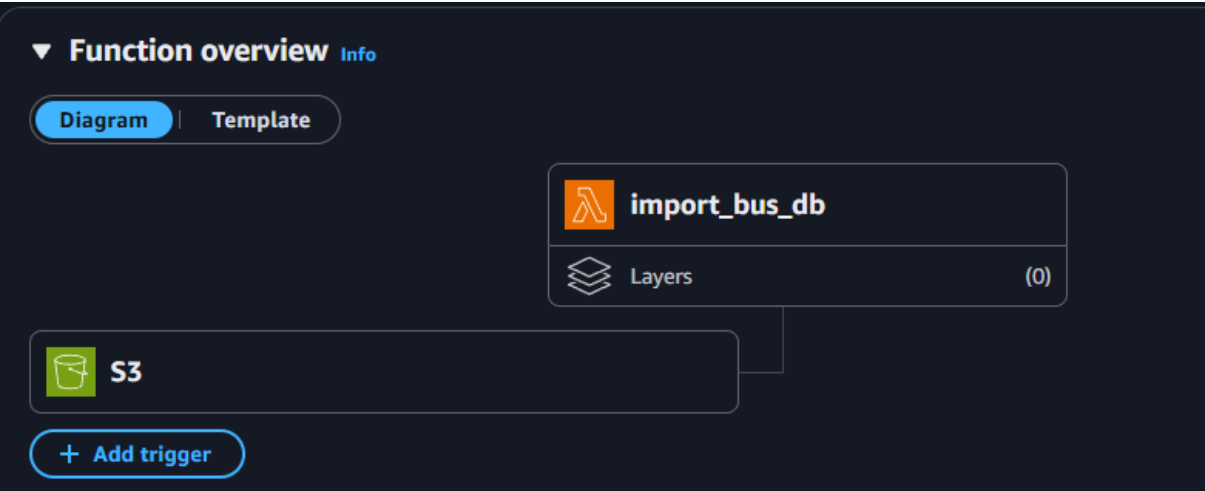
S3
aws asynchronous storage

Bucket
Choose or enter the ARN of an S3 bucket that serves as the event source. The bucket must be in the same region as the function.
s3/bus-db-data
Bucket region: eu-south-2

Event types
Select the events that you want to have trigger the Lambda function. You can optionally set up a prefix or suffix for an event. However, for each bucket, individual events cannot have multiple configurations.
All object create events X PUT X

Prefix - optional
Enter a single optional prefix to limit the notifications to objects with keys that start with matching characters. Any special characters must be URL encoded.
e.g. images/

Suffix - optional
Enter a single optional suffix to limit the notifications to objects with keys that end with matching characters. Any special characters must be URL encoded.
.json



During this process several issues appeared, and both the CloudWatch service and different metrics (both in Lambda and CloudWatch) were very useful to fix errors

CloudWatch > Log management > /aws/lambda/import_bus_db > 2026/05/24/[\$LATEST]504e57f30d10444

CloudWatch

Favorites and recents

Ingestion

Dashboards

Alarms  0  0  0

► AI Operations

► Application Signals (APM)

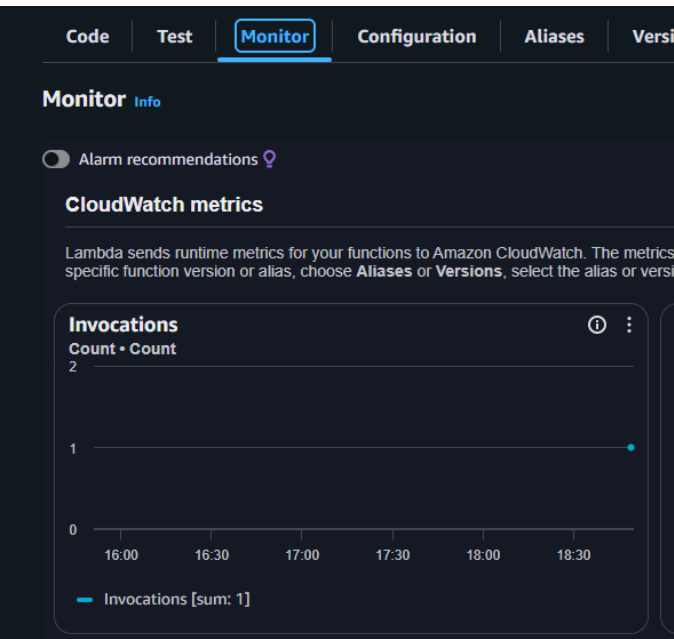
► Infrastructure Monitoring

Log events

You can use the filter bar below to search for and match terms, phrases, or values in

🔍 Filter events - press enter to search

▶	Timestamp	Message
		No older events at this moment. View all
▶	2026-05-24T18:49:08.868Z	INIT_START Runtime Version: python3.11
▶	2026-05-24T18:49:09.131Z	START RequestId: cca6c63f-75d5-4b8a-b0e1-1a1a1a1a1a1a
▶	2026-05-24T18:49:09.132Z	[ERROR] NameError: name 'boto3' is not defined



- One of the errors seemed to be caused by default time limitations in lambda method execution, which was increased.

Edit basic settings

Basic settings

Description - optional

Memory

Your function is allocated CPU proportional to the memory configured.

166

MB

Set memory to between 128 MB and 10240 MB

Ephemeral storage

You can configure up to 10 GB of ephemeral storage (/tmp) for your function. [View pricing](#)

512

MB

Set ephemeral storage (/tmp) to between 512 MB and 10240 MB.

SnapStart

Reduce startup time by having Lambda cache a snapshot of your function after the function has initialized. To evaluate whether your function code is resilient to snapshot operation [pricing](#).

None

Supported runtimes: .NET 10 (C#/F#/PowerShell), .NET 8 (C#/F#/PowerShell), Java 11, Java 17, Java 21, Java 25, Python 3.12, Python 3.13, Python 3.14.

Timeout

0

min

38

sec

Execution role

To access other AWS services and resources your function needs an IAM role. Choose a role with the right permissions for your function.

import-from-lambda-role

Create new role

[View role details in IAM](#)

- Once errors with settings and code were fixed, database was properly updated

Table: BusData - Items returned (36)

Scan started on May 24, 2026, 23:55:25

< 1 >

☐	PK (String)	SK (String)	dataset_id	destination	route	stop_name	test	time
☐	DEBUG	CHECK					yes	
☐	STOP#171	TIME#09:55	2026-05-24...	East Bus Station	5			
☐	STOP#171	TIME#19:55	2026-05-24... 📄 🔗	East Bus Station	5			
☐	STOP#42	TIME#19:25	2026-05-24...	St George Ho...	6			
☐	STOP#196	TIME#08:40	2026-05-24...	Airport	2			
☐	STOP#196	TIME#18:45	2026-05-24...	St George Ho...	6			
☐	STOP#111	TIME#08:00	2026-05-24...	Airport	2			
☐	TRIP#5_0001	STOP#132	2026-05-24...			Park Ave		09:12
☐	TRIP#5_0001	STOP#162	2026-05-24...			Market Squ...		09:28
☐	TRIP#5_0001	STOP#168	2026-05-24...			East Campus		09:45
☐	TRIP#5_0001	STOP#171	2026-05-24...			East Bus St...		09:55
☐	CONFIG	ACTIVE_DATASET	2026-05-24...					

The other lambda functions were created in a similar way

Author from scratch
Start with a simple Hello World example.

Use a blueprint
Build a Lambda application from sample code and configuration presets for common use cases.

Container image
Select a container image to deploy for your function.

Basic information

Function name
Enter a name that describes the purpose of your function.

Function name must be 1 to 64 characters, must be unique to the Region, and can't include spaces. Valid characters are a-z, A-Z, 0-9, hyphens (-), and underscores (_).

Runtime [Info](#)
Choose the language to use to write your function. Note that the console code editor supports only Node.js, Python, and Ruby.

Python 3.14

Permissions
By default, Lambda will create an execution role with permissions to upload logs to Amazon CloudWatch Logs. To use a different role, choose **Custom execution** role under **More settings**.

Custom settings [Info](#)

Durable execution - new
EC2 capacity provider - new

Execution role
To access other AWS services and resources your function needs, choose a role with the right permissions for your function.

Choose an existing role

Lambda function service-assumable roles

- import-from-lambda-role
- query_bus_db_from_lambda_role

Additional roles

- AWSServiceRoleForResourceExplorer
- AWSServiceRoleForSupport
- AWSServiceRoleForTrustedAdvisor
- ImageGeneratorBucketUploader
- TestRoleEC2

And all could be tested in “Test” section, with custom inputs:

Event name
testQuery1
Maximum of 25 characters consisting of letters, numbers, dots, hyphens and underscores.

Event sharing settings
☒ Private
This event is only available in the Lambda console and to the event creator. You can configure a total of 10. [Learn more](#)
☐ Shareable
This event is available to IAM users within the same account who have permissions to access and use shareable events. [Learn more](#)

Template - optional
Hello World

Event JSON

```

1 {
2   "body": "{\"stop_id\":\"111\"}"
3 }

```

[Format JSON](#)

Code
Test
Monitor
Configuration
Aliases
Versions

Executing function: succeeded (logs)

Details

```

{
  "statusCode": 200,
  "body": "{\"stop_id\": \"111\", \"count\": 1, \"items\": [{\"dataset_id\": \"2026-05-24T21:48:24Z\", \"destination\": \"Airport\", \"trip_id\": \"2_0001\", \"PK\": \"STOP#111\", \"route\": \"2\", \"SK\": \"TIME#00:00\"}]}"
}

```

Summary
Code SHA-256: RplToWqgvC...IGCs=

Details

```

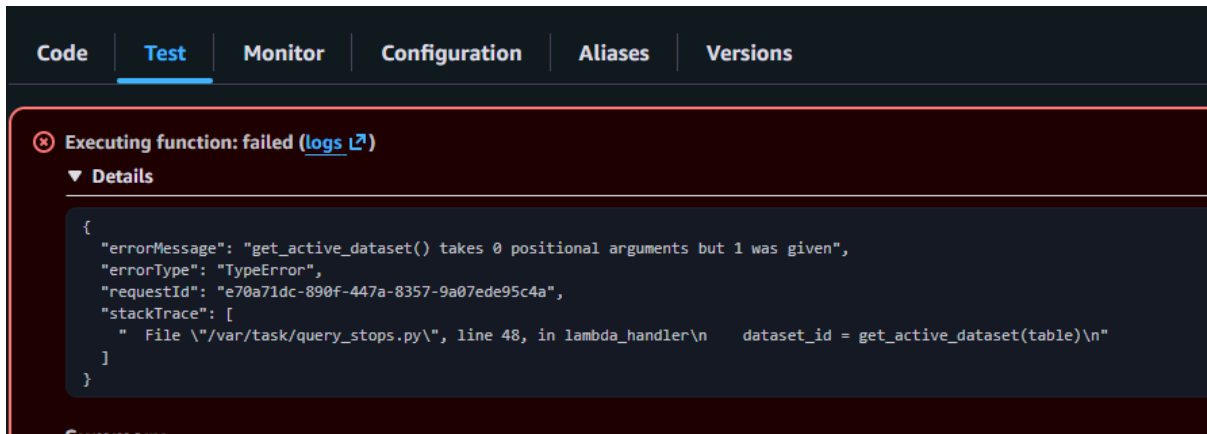
{
  "statusCode": 200,
  "body": "{\"trip_id\": \"2_0001\", \"count\": 5, \"items\": [{\"dataset_id\": \"2026-05-24T21:48:24Z\", \"time\": \"08:00\", \"PK\": \"TRIP#2_0001\", \"stop_name\": \"Main St\", \"SK\": \"STOP#111\", {\"dataset_id\": \"2026-05-24T21:48:24Z\", \"time\": \"08:10\", \"PK\": \"TRIP#2_0001\", \"stop_name\": \"Park Ave\", \"SK\": \"STOP#131\", {\"dataset_id\": \"2026-05-24T21:48:24Z\", \"time\": \"08:20\", \"PK\": \"TRIP#2_0001\", \"stop_name\": \"St John Square\", \"SK\": \"STOP#152\", {\"dataset_id\": \"2026-05-24T21:48:24Z\", \"time\": \"08:40\", \"PK\": \"TRIP#2_0001\", \"stop_name\": \"Airport\", \"SK\": \"STOP#196\"}]}}]"
}

```

Summary

Code SHA-256 RplToWqgvC...IGCs=	Execution time 21 seconds ago
Function version \$LATEST	Request ID 4d60aa78-7448-4eff-95f0-968a812c4131
Duration 355.26 ms	Billed duration 831 ms
Resources configured 128 MB	Max memory used 93 MB
Init duration 475.23 ms	

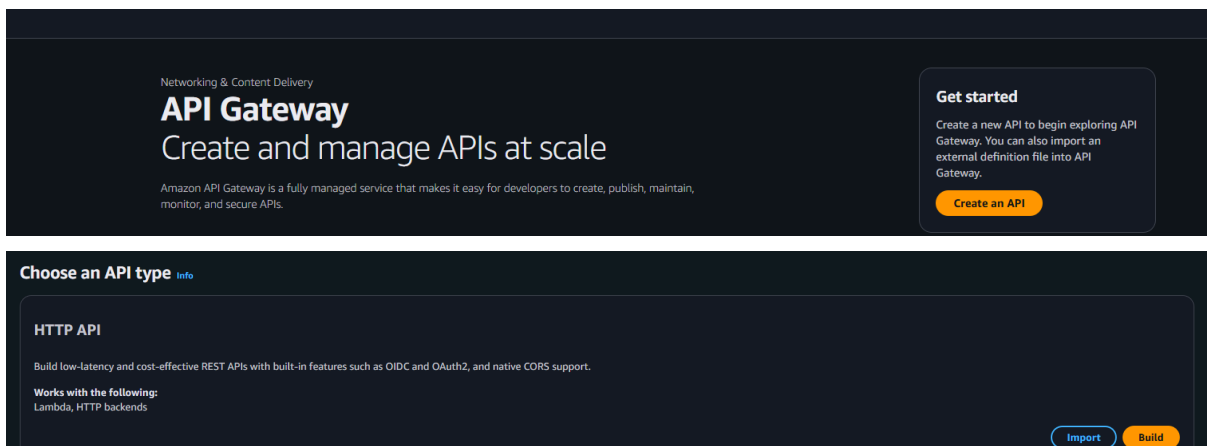
And detect errors with code



API Gateway:

For a simple and cheap option, with no need for cache, HTTP API was chosen as a base for the querying service

- We select create an API, and then select HTTP :



- We select the lambdas

Configure API

API details

API name
An HTTP API must have a name. The name is a non-unique value you use to identify and organize your APIs. To programmatically refer to this API, use the API ID that API Gateway generates for you.

bus-db-api

IP address type [Info](#)
Select the type of IP addresses that can invoke the default endpoint for your API. You don't need to redeploy your API for the update to take effect.

☒ **IPv4**
Includes only IPv4 addresses.

☐ **Dualstack**
Includes IPv4 and IPv6 addresses.

Integrations (2) [Info](#)

Specify the backend services that your API will communicate with. These are called integrations. For a Lambda integration, API Gateway invokes the Lambda function and responds with the function's response. For an HTTP integration, API Gateway sends the request to the URL that you specify and returns the response from the URL.

Lambda

AWS Region eu-south-2 **Lambda function** **Version** 2.0 [Learn more](#) [Remove](#)

Lambda

AWS Region eu-south-2 **Lambda function** **Version** 2.0 [Learn more](#) [Remove](#)

- And we configure routes. Then we simply accept and create the service.

Configure routes - optional

Configure routes [Info](#)
API Gateway uses routes to expose integrations to consumers of your API. Routes for HTTP APIs consist of two parts: an HTTP method and a resource path (e.g., GET /pets). You can define specific HTTP methods for your integration (GET, POST, PUT, PATCH, HEAD, OPTIONS, and DELETE) or use the ANY method to match all methods that you haven't defined on a given resource.

Method	Resource path	Integration target	
GET	/stops/{trip_id}	query_bus_db_by_trip	Remove
GET	/trips/{stop_id}	query_bus_db_by_stop	Remove

[Add route](#)

API Gateway

APIs
Custom domain names
Domain name access associations
VPC links

API: bus-db-api-424

Develop
Routes
Authorization
Integrations
CORS
Reimport
Export

Deploy
Stages

bus-db-api

Stage: - [Deploy](#)

API details [Edit](#)

API ID 424 **Protocol** HTTP **Created** 2026-05-26

Description - **IP address type** IPv4 **Default endpoint** [Enabled](#) [https://execute-api-eu-south-2.amazonaws.com](#)

ARN [arn:aws:apigateway:eu-south-2::apis/424](#)

Stages for bus-db-api (1)

Stage name	Invoke URL	Attached deployment	Auto deploy	Last updated
\$default	https://execute-api-eu-south-2.amazonaws.com		enabled	2026-05-26

- Finally, for security reasons, we limit the amount of requests the API will handle. For a simple learning project low numbers were chosen. Since all information is public, and no part of the service is based on specific users or accounts, design was kept simple and auth was omitted.

API Gateway

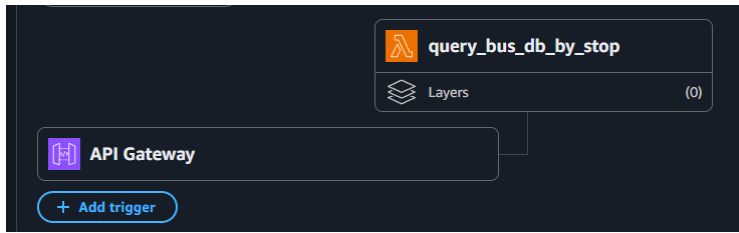
APIs
Custom domain names
Domain name access associations
VPC links

Throttling for \$default stage

Edit default route throttling

Burst limit **Rate limit**

If we check the lambda functions used for querying, we can now see that there are new triggers



Also, since inputs coming from API Gateway had a different structure, code in lambda needed minor adjustments. After some changes, those functions could handle different kinds of inputs.